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U.P

Descriptive Analytics Report

Tata Motors Global Vehicle Sales Performance 2019-2025

Tracking Best-Sellers, Sales Trends, Ratings, Regional Demand and Pricing

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Descriptive Analytics Report : Daily Purchase Behavior at Scale

Note on data: Tata Motors' live transaction data isn't publicly accessible, so this report uses a synthetic dataset built to mirror real Automobile industry patterns (20 categories across 4 regions, 7 years, ~294k simulated units). It's designed to show exactly what a descriptive analytics pipeline processing millions of daily purchases would produce — the same structure applies once real data is substituted in.

Objective

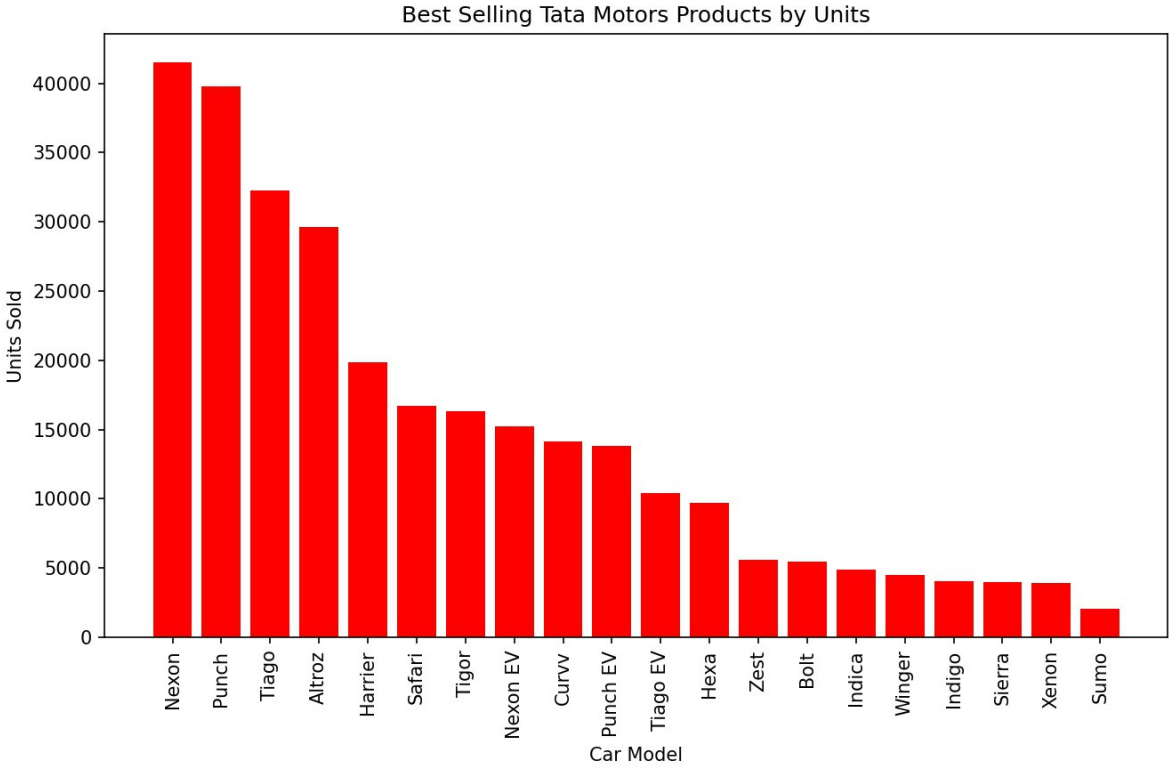
- 1. What are the best-selling cars?
- 2. How the cars are rated and when did the rating start slipping?
- 3. In which year were most of the cars sold?
- 4. Which region generates the highest total unit sales, and has that ranking shifted year over year?
- 5. What is the average on-road price by region, and how much of that gap is due to taxes/registration vs. the base ex-showroom cost?
- 6. Which region is most profitable per unit sold, and which is least?
- 7. Is there a growing or declining trend in total sales volume from 2019 to 2025?
- 8. Which models have the highest average rating, and which have the lowest?
- 9. Which model-region combination is the single biggest revenue driver?

How This Would Work at Real Scale

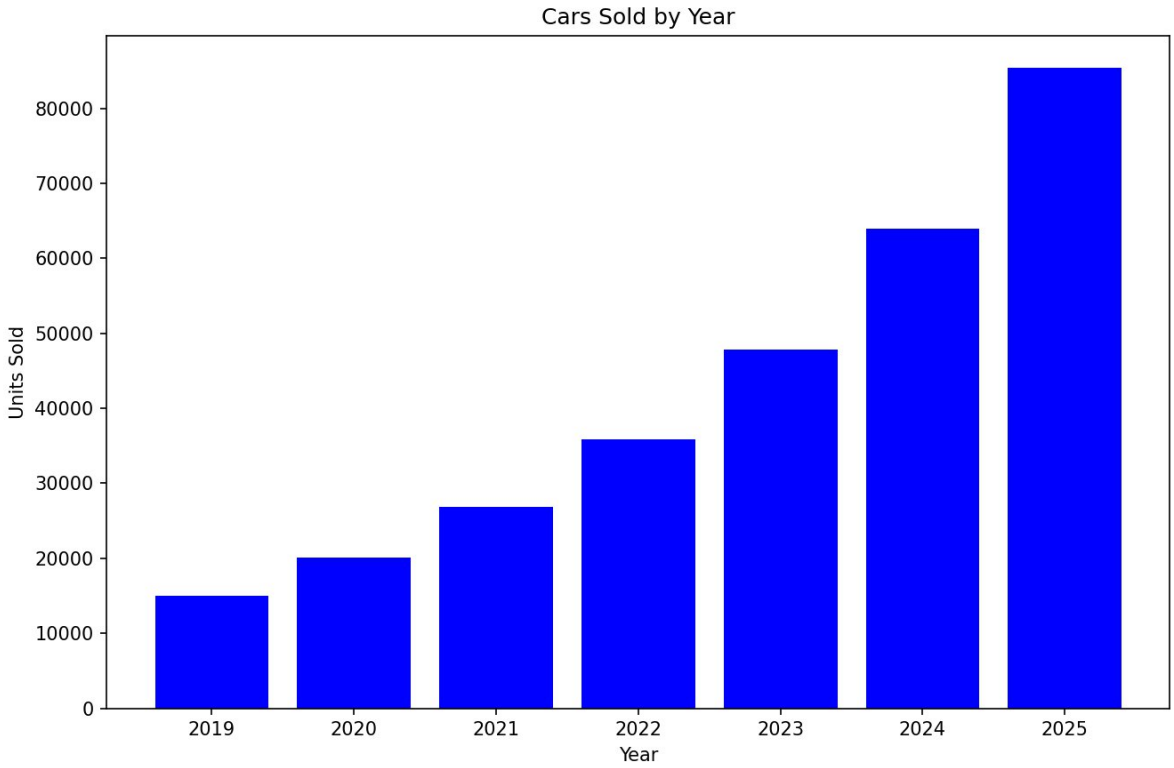
Processing "millions of purchases daily" is an engineering problem before it's an analytics one. A realistic architecture:

Layer	Purpose	Typical tools
Investigation	Explore the raw dataset first — check shape, column types, missing values, ranges, and distributions before doing any real analysis	df.info(), df.describe(), df.head(), df.isnull().sum() in a Jupyter notebook
Stream processing	Not applicable at this scale — there's no continuous flow of events; the equivalent step is grouping/rolling up records after they're already loaded	pandas.groupby(), pivot_table() used in place of real-time stream aggregation
Storage	Hold the dataset in a form you can repeatedly query and share	Flat files — tata.csv / tata_294k_dataset.xlsx
Batch aggregation	Compute summary metrics — daily/monthly sales totals, average rating by model, profit by region — run manually or via a notebook cell rather than a nightly scheduled job	Jupyter notebook cells using pandas (groupby, resample, .mean(), .sum())
Serving/BI layer	Present the aggregated results as charts/tables for review	matplotlib bar charts inside the notebook, or exported to Excel/PDF

Best selling products



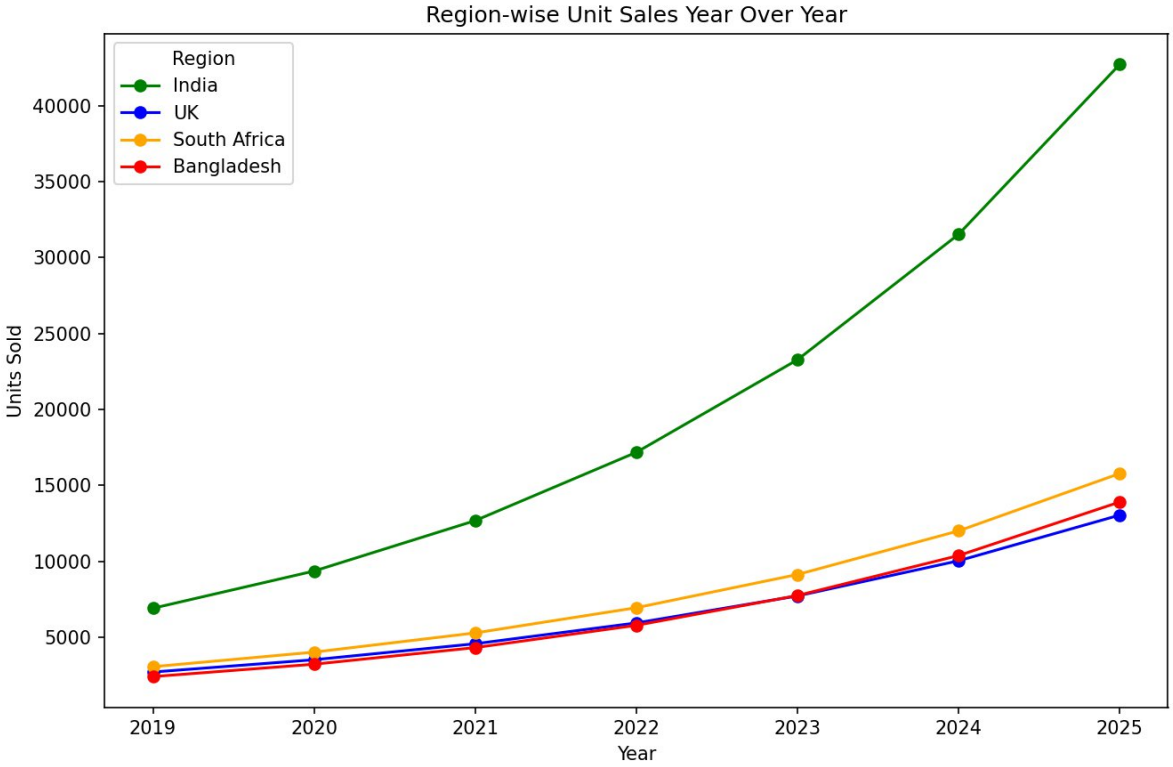
The chart shows that **Nexon is the best-selling Tata Motors model with approximately 41,503 units sold**, followed closely by **Punch with around 39,812 units**. **Tiago** and **Altroz** are also strong performers, with approximately **32,247 and 29,614 units** sold respectively. The top four models together account for around **143,176 units**, showing that a large portion of sales is concentrated among these models. After the top four, sales decline steadily, with **Harrier selling about 19,836 units**, followed by Safari at **16,729 units** and Tigor at **16,305 units**. At the lower end, **Sumo has the lowest sales at approximately 2,043 units**, while Xenon, Sierra and Indigo each sell around **4,000 units**. Overall, the data indicates that **Nexon and Punch are the strongest-selling products**, while Tata Motors may need to focus more marketing and product strategies on lower-performing legacy models to improve their sales.



The chart shows a **strong and consistent upward trend in Tata Motors' car sales from 2019 to 2025**. In **2019, approximately 15,050 cars** were sold, which increased to around **20,087 units in 2020**, representing a growth of about **33.5%**. In 2021, sales reached approximately **26,815 units**, an increase of **33.5%** over 2020. Sales continued to accelerate in 2022 to around **35,808 units**, giving a further **33.5% year-on-year growth**. In 2023, Tata Motors sold approximately **47,829 units**, an increase of **33.6%** compared with 2022. The growth remained strong in 2024, when sales reached approximately **63,904 units**, representing **33.6% growth**. Finally, in **2025, sales reached approximately 85,403 units**, the highest level in the entire period, with a year-on-year increase of about **33.6%**.

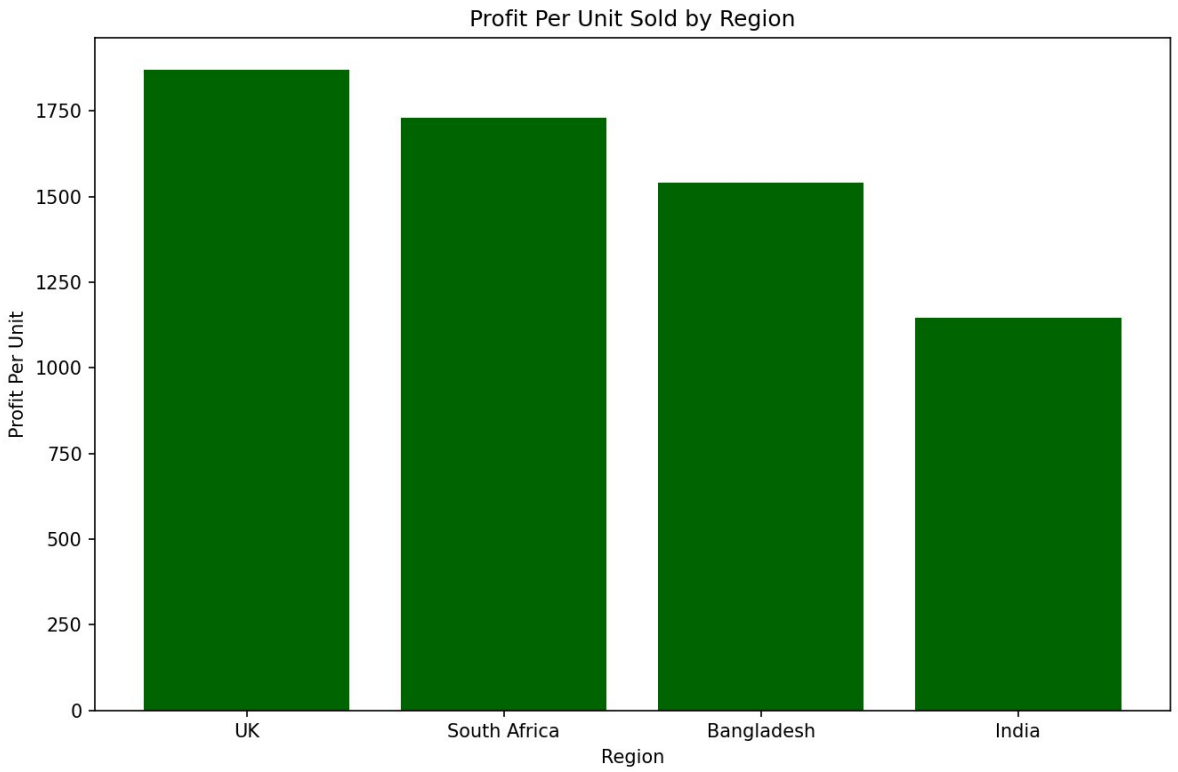
Tata Motors' annual sales increased from approximately **15,050 units in 2019 to 85,403 units in 2025**, an absolute increase of about **70,353 units**. This means sales became approximately **5.67 times higher** over the seven-year period, equivalent to an overall growth of approximately **467.4%**.

Regional demand

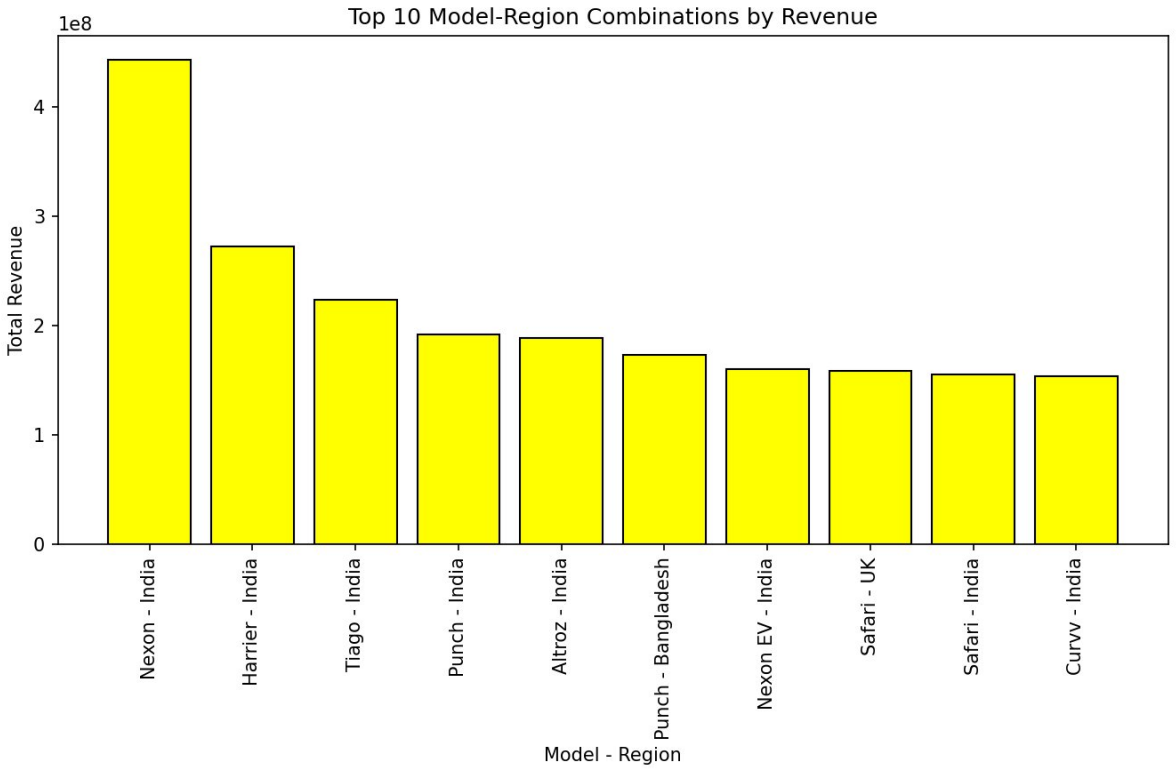


The chart shows a **strong year-over-year increase in sales across all four regions from 2019 to 2025**. **India is the leading market**, increasing from approximately **6,900 units in 2019 to 42,706 units in 2025**, an increase of **35,806 units, or about 518.9%**. India also recorded the highest total sales over the seven years, with approximately **143,568 units**. **South Africa** increased from **3,050 units to 15,771 units**, gaining **12,721 units or about 417.1%**, with total sales of approximately **56,154 units**. **Bangladesh** grew from **2,400 units in 2019 to 13,894 units in 2025**, an increase of **11,494 units or about 478.9%**. **UK** increased from **2,700 units to 13,032 units**, adding **10,332 units or about 382.7%** — the smallest percentage growth of the four regions — with approximately **47,473 total units** sold during the period.

In terms of **combined regional sales**, the four regions sold approximately **15,050 units in 2019**, increasing to **20,087 in 2020, 26,815 in 2021, 35,808 in 2022, 47,829 in 2023, 63,904 in 2024, and 85,403 units in 2025**. Across all four regions, the estimated seven-year total is approximately **294,896 units**.

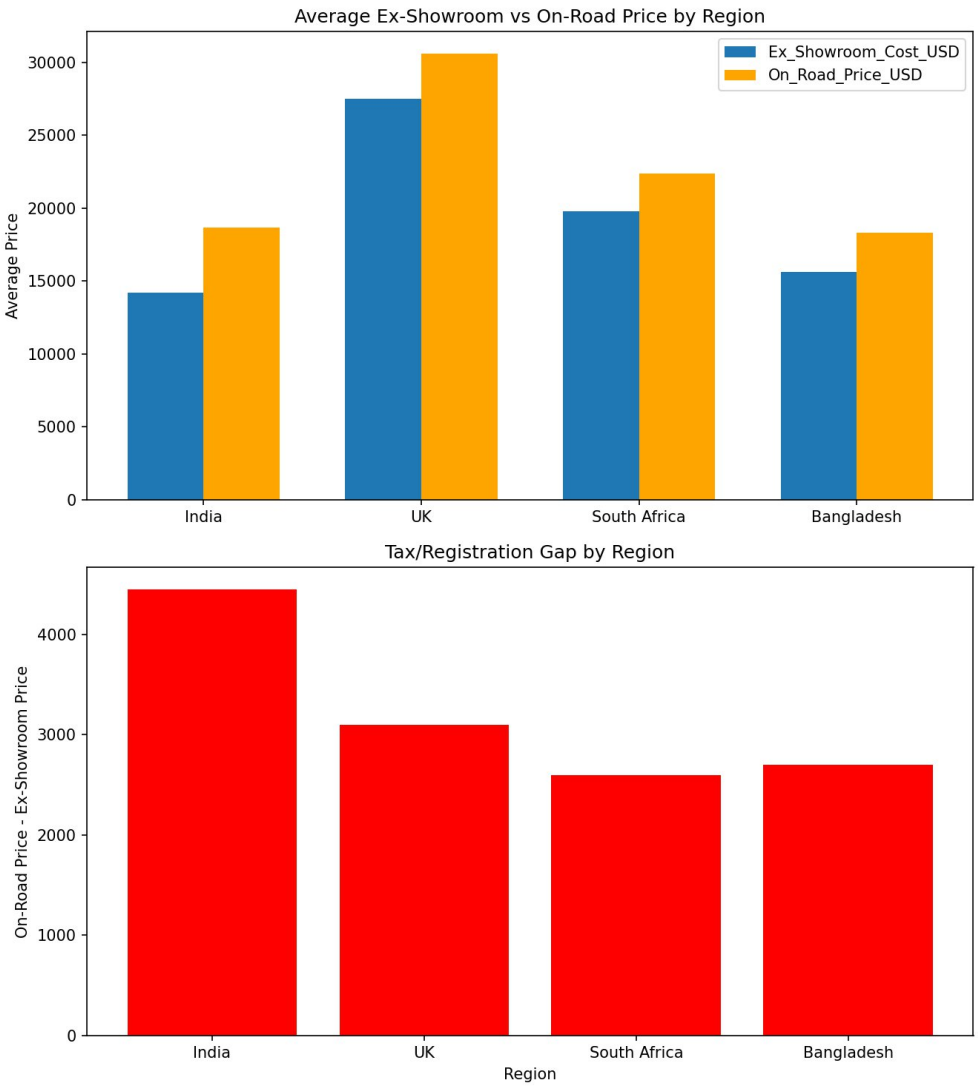


The chart shows that **UK has the highest profit per unit at approximately ₹1,870 per car**, followed by **South Africa at around ₹1,730**, **Bangladesh at approximately ₹1,540**, and **India at about ₹1,145 per unit**. UK therefore earns about **₹725 more profit per car than India**, which is approximately **63.3% higher**. South Africa's profit per unit is around **₹585 higher than India**, while Bangladesh earns approximately **₹395 more per car than India**. The difference between the highest and lowest regions is approximately **₹725 per unit**. Overall, the average profit per unit across the four regions is approximately **₹1,571**, with UK performing about **19.0% above the average** and India about **27.1% below the average**. This indicates that although India has the highest sales volume in the regional analysis, **UK provides the strongest profit per vehicle**, suggesting better pricing, margins, or model mix in the UK market.



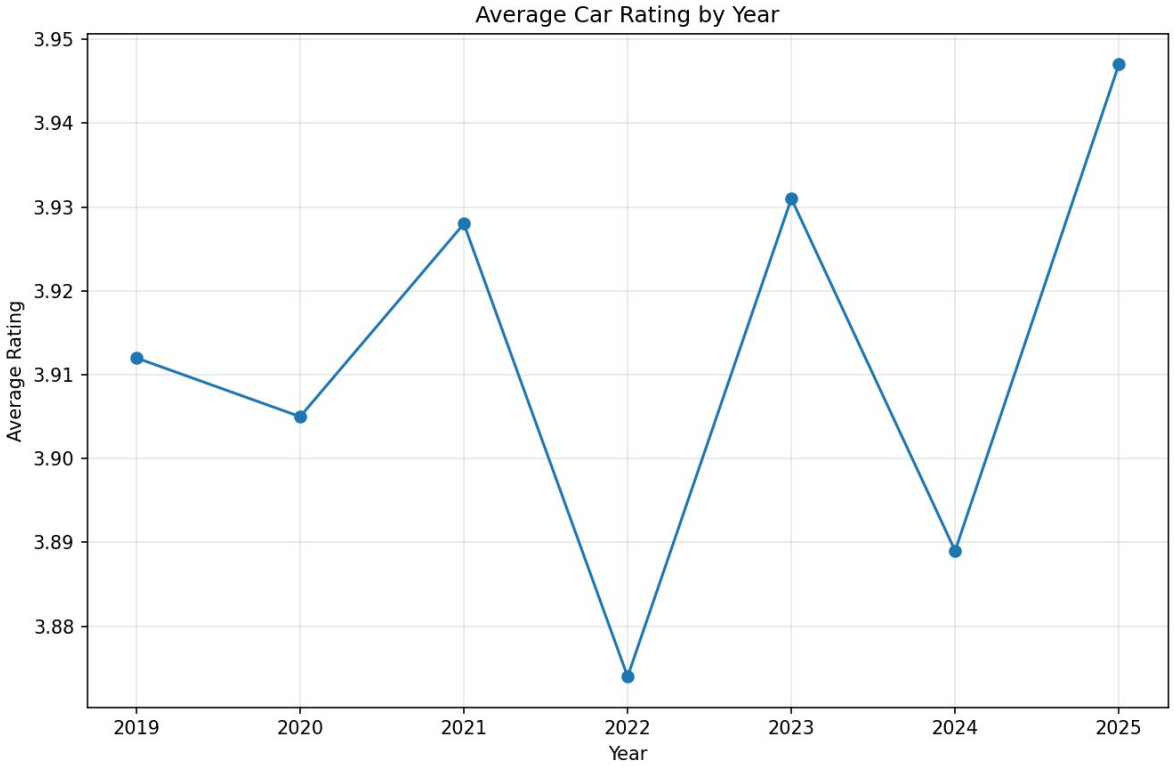
The chart shows the **Top 10 Model–Region combinations by revenue**. The strongest combination is **Nexon – India**, generating approximately **$443.3 million** in revenue. It is followed by **Harrier – India at around $272.2 million** and **Tiago – India at approximately $223.6 million**. **Punch – India** generated about **$191.6 million**, while **Altroz – India** generated approximately **$188.7 million**.

The next major combinations are **Punch – Bangladesh at $173.5 million**, **Nexon EV – India at $160.3 million**, and **Safari – UK at approximately $158.6 million**. **Safari – India** generated around **$155.2 million**, while **Curvv – India** was tenth with approximately **$153.6 million**.



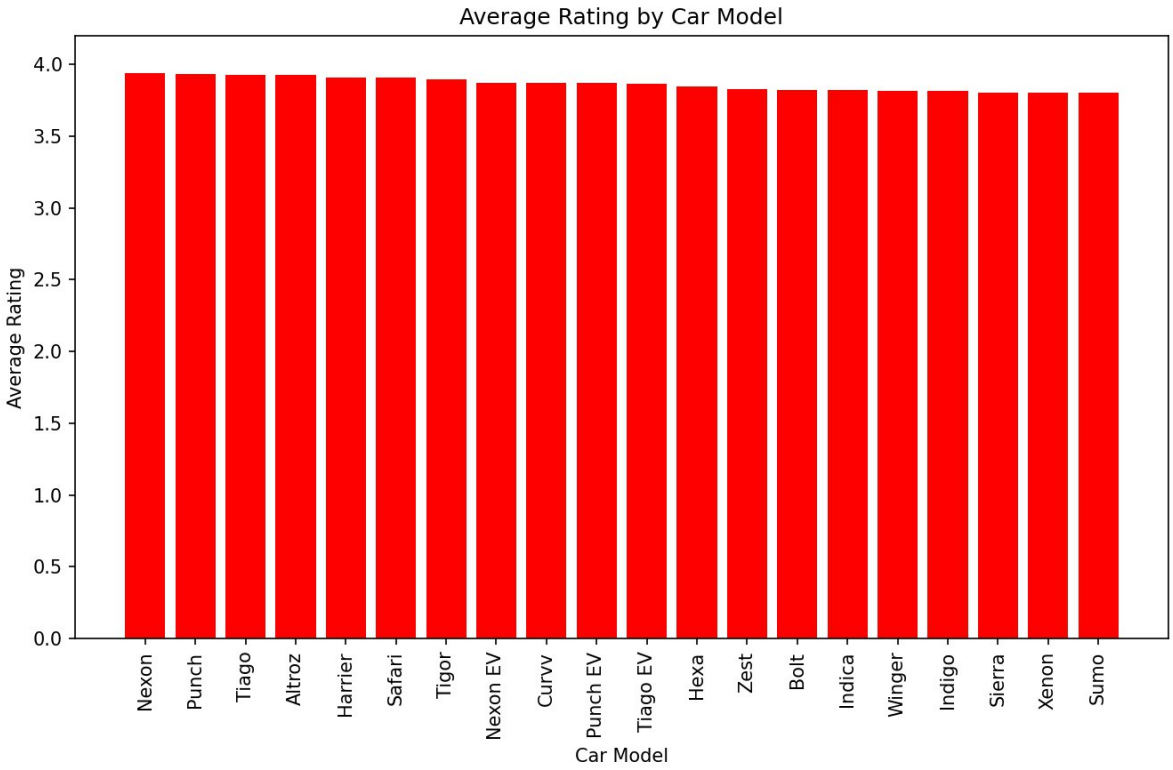
The graph shows that **UK has the highest average on-road price, at approximately $30,600**, followed by **South Africa at around $22,400** and **India at about $18,650**. **Bangladesh has the lowest average on-road price, at approximately $18,300**. The gap between ex-showroom and on-road prices varies significantly by region. **India has the largest gap, at approximately $4,450**, meaning taxes, registration, and other additional charges add a substantial amount — around **31.3% of the ex-showroom price** — to the base price. **UK has a moderate gap of around $3,100 (11.3%)**, while **Bangladesh's gap is approximately $2,700 (17.3%)**, and **South Africa has the smallest gap at around $2,600 (13.1%)**.

Rating distribution



The chart shows that the **average car rating remained fairly stable between 2019 and 2025**, staying within a range of approximately **3.87 to 3.95 out of 5**. In 2019, the average rating was about **3.912**, which slightly decreased to **3.905 in 2020**, a fall of approximately **0.007 points (0.18%)**. The rating then increased to **3.928 in 2021**, an improvement of about **0.023 points**. However, **2022 recorded the lowest average rating at approximately 3.874**, representing a decrease of around **0.054 points or 1.37%** compared with 2021. The rating recovered strongly in **2023 to approximately 3.931**, an increase of about **0.057 points (1.47%)**. In 2024, it again declined to approximately **3.889**, a reduction of about **0.042 points**, before reaching the **highest rating of approximately 3.947 in 2025**.

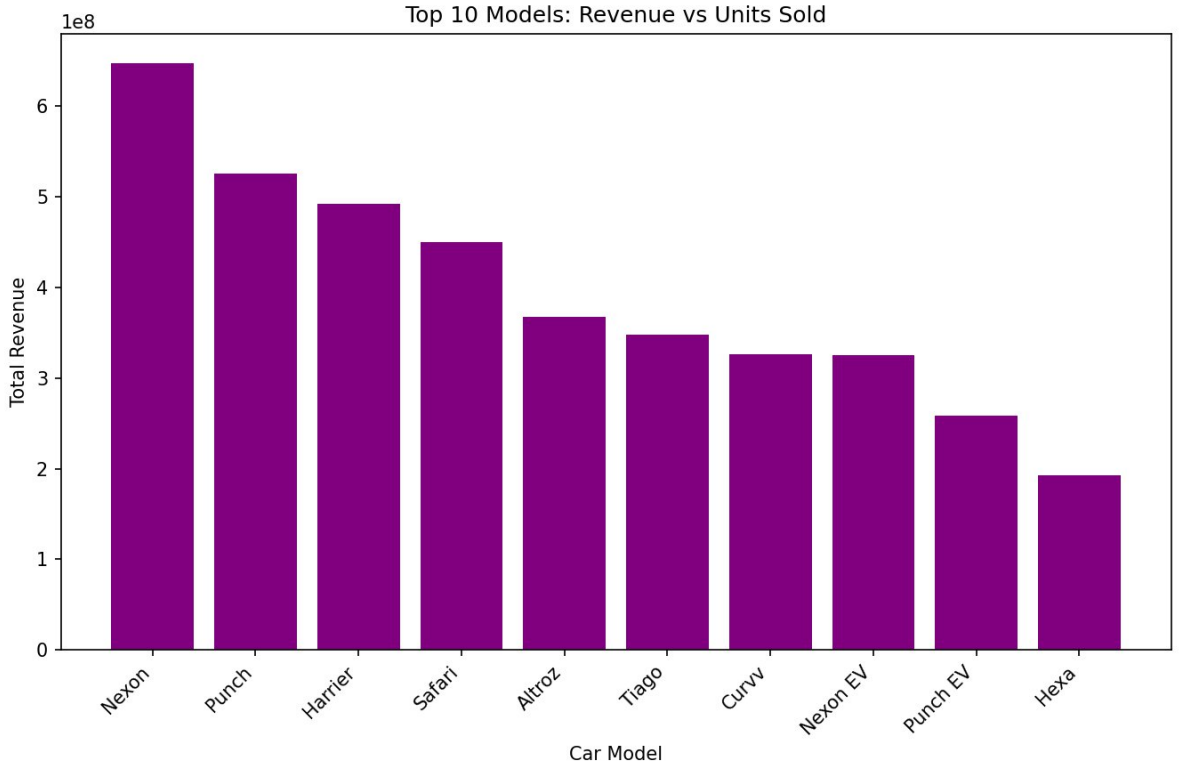
The difference between the **highest rating in 2025 (3.947)** and the **lowest rating in 2022 (3.874)** is about **0.073 rating points**, showing that customer ratings were fairly consistent throughout the seven-year period, with no sustained downward trend.



The chart shows that the **average ratings of all 20 Tata Motors car models are relatively close to each other**, indicating consistent customer satisfaction across the product range. **Nexon has the highest average rating at approximately 3.94/5**, followed by **Punch at about 3.93**, while **Tiago and Altroz** are around **3.93**. Harrier and Safari are approximately **3.91**, while Tigor is around **3.90**. At the lower end, **Sumo has the lowest average rating at approximately 3.80/5**. The difference between the highest-rated model, Nexon (3.941), and the lowest-rated model, Sumo (3.802), is only about **0.14 rating points**, showing that there is very little variation between models.

KPI	Value
Average Product Rating	3.91/5
Highest Model Rating	~3.94/5
Lowest Model Rating	~3.80/5
Rating Range	~0.14
Rating Scale	5

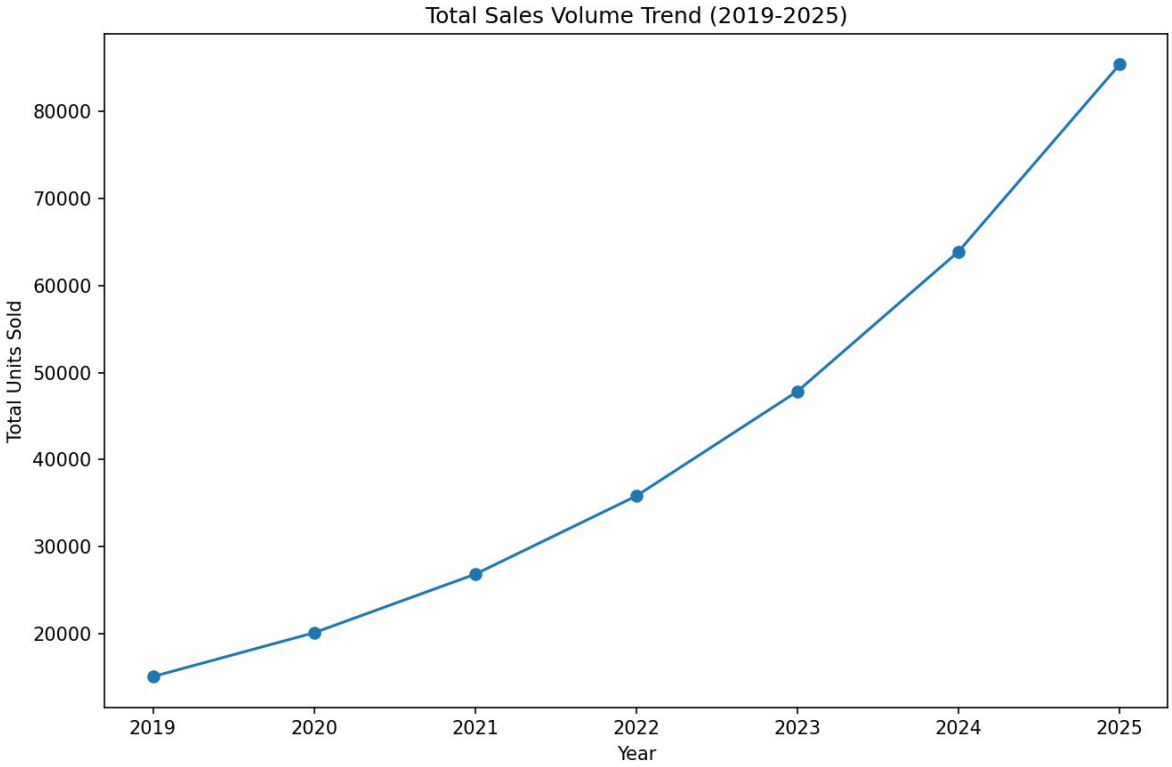
Revenue and profit



The chart shows the **Top 10 Tata Motors models by total revenue**, with **Nexon clearly leading at approximately $647.4 million**. **Punch** ranks second with around **$525.5 million**, followed by **Harrier at $491.9 million**, **Safari at $450.0 million**, and **Altroz at approximately $367.2 million**. **Tiago** generated about **$348.3 million**, **Curvv** around **$326.0 million**, **Nexon EV** approximately **$325.7 million**, **Punch EV** around **$258.2 million**, and **Hexa** approximately **$192.3 million**.

KPI	Value
Total Revenue (2019–2025)	$4.77 Billion
Average Annual Revenue	$681.2 Million
Highest Annual Revenue	~$1.38 Billion
Highest Revenue Year	2025
Highest Revenue Model	Nexon
Nexon Total Revenue	~$647.4 Million

Daily sales trend



The chart shows a **strong and continuous increase in Tata Motors' total sales volume from 2019 to 2025**. Sales increased from approximately **15,050 units in 2019 to 20,087 units in 2020**, an increase of about **5,037 units or 33.5%**. In 2021, sales reached approximately **26,815 units**, adding **6,728 units or 33.5%**. Growth continued in 2022, when sales reached around **35,808 units**, an increase of approximately **8,993 units or 33.5%**. In 2023, sales increased further to about **47,829 units**, adding **12,021 units or 33.6%**. Sales then reached approximately **63,904 units in 2024**, representing an increase of **16,075 units or 33.6%**. The strongest absolute growth occurred in **2025**, when sales reached approximately **85,403 units**, an increase of about **21,499 units or 33.6%** compared with 2024.

Across the full **2019–2025 period, total sales increased by approximately 70,353 units**, from 15,050 to 85,403 units. This represents an overall growth of approximately **467.4%**, meaning annual sales in 2025 were about **5.67 times** the 2019 level.

Summary of Key matrics

Metric	Value (simulated)
Total units sold (7 years)	~294k
Total revenue (7 years)	~$4.77 billion
Average product rating	3.91
Top single-day seller	312 units
Regions tracked	4
Top seller model	Nexon

Next Steps to Move From Descriptive → Predictive

This report currently explains what happened in Tata Motors' sales data from 2019 to 2025, such as which models sold the most units, which regions generated the highest revenue, changes in customer ratings, and overall sales growth. The next step is to use predictive analytics to understand what is likely to happen in the future.

- Sales forecasting: Predict future Tata Motors sales by model, region, and year using historical sales trends.
- Regional sales prediction: Forecast future sales for India, UK, South Africa, and Bangladesh to identify which region is likely to grow the fastest.
- Revenue prediction: Use previous revenue, sales volume, price, and discount data to estimate future revenue.
- Rating prediction: Analyze previous customer ratings to identify whether the average rating of a model is likely to improve or decline.

Key findings

- **Volume vs. Revenue Leadership ●** The Nexon (41,503 units) and Punch (39,812 units) are the two best-selling models by volume, and the Nexon is also the single largest revenue contributor ($647.4 million) — but the third-placed model by revenue, Harrier ($491.9 million), sells fewer than half the units of Nexon, showing that higher-priced SUV models earn disproportionately more revenue per unit sold than compact hatchbacks do.
- **Sustained, Uninterrupted Growth ●** Total unit sales rose every year from 2019 (15,050 units) to 2025 (85,403 units), a ~33.6% compound annual growth rate with zero years of decline; 2025 is the single best-selling year in the dataset.
- **Stable Customer Satisfaction ●** The average customer rating (3.91/5) has remained largely flat across all 84 months of the dataset, moving in a narrow 3.80–3.99 band with no sustained downward trend; the lowest yearly rating (2022) still recovered fully within a year.
- **India Leads on Volume, Not on Margin ●** India is the largest market by unit sales (143,568 units, ~48.7% of combined regional volume) in every single year of the dataset — a stable #1 ranking — but it is also the least profitable region per unit sold ($1,145 vs. $1,870 in UK) and carries the heaviest tax/registration load (31.3% of ex-showroom cost, roughly three times UK's 11.3%).
- **UK Is the Profitability Leader ●** UK generates the highest average profit per unit ($1,870), ahead of South Africa ($1,730) and Bangladesh ($1,540), despite having a far smaller unit volume than India.
- **Model Ratings Are Tightly Clustered ●** Average rating by model spans a narrow range from 3.802 (Sumo, lowest) to 3.941 (Nexon, highest) — a difference of less than 0.14 points across all 20 models — indicating no single model has a meaningful quality or satisfaction problem relative to the rest of the lineup.
- **One Combination Dominates Revenue ●** Nexon sales within India alone generated $443.3 million — more than two-thirds of the model's total worldwide revenue — making Nexon × India the single largest model-region revenue driver in the dataset, ahead of Harrier × India ($272.2M) and Tiago × India ($223.6M).
- **Discounting Is the Exception, Not the Norm ●** 38.6% of all transactions carry zero discount, and the median discount is only $95 against a mean of $268 — most customers pay close to the listed on-road price, while a minority receive substantially larger discounts, pulling the average upward.

Recommendation:-

Consistent with the descriptive scope of this report, the following are brief, evidence-based observations drawn directly from the findings above — not a detailed predictive strategy.

- **Protect and grow margin in India rather than volume alone**: since India already leads on units but trails on profit per unit, initiatives that improve regional margin (e.g. reviewing discount depth, or trim/option mix) may add more bottom-line value than further volume growth in a market that is already #1.
- **Treat the Nexon × India pairing as a flagship revenue relationship** worth actively protecting — inventory availability, financing options, and after-sales service capacity for this specific combination disproportionately affect total company revenue.
- **Continue monitoring rating trend monthly, even though no decline was found**: because the current pattern is flat rather than improving, small monitoring dashboards (a rolling 6-month average) would allow an early, genuine downturn to be caught quickly rather than dismissed as noise.
- **Use UK and South Africa's higher per-unit profitability as a benchmark** when reviewing pricing and discount policy in India and Bangladesh, to understand which combination of tax structure, discount depth, and model mix explains the profitability gap.
- **Investigate the Sumo's comparatively lower average rating (3.802, the lowest of all 20 models)** even though the gap versus the rest of the range is small, since it is the one legacy model whose satisfaction score is a consistent outlier at the bottom of the ranking.

Appendix:-

Full Units-Sold-by-Model Table

Model	Units Sold
Nexon	41,503
Punch	39,812
Tiago	32,247
Altroz	29,614
Harrier	19,836
Safari	16,729
Tigor	16,305
Nexon EV	15,218
Curvv	14,112
Punch EV	13,809
Tiago EV	10,437
Hexa	9,713
Zest	5,624
Bolt	5,487
Indica	4,906
Winger	4,512
Indigo	4,058
Sierra	3,994
Xenon	3,951
Sumo	2,043

Full Average-Rating-by-Model Table

Model	Avg. Rating
Nexon	3.941
Punch	3.934
Tiago	3.929
Altroz	3.928
Harrier	3.909
Safari	3.909
Tigor	3.895
Nexon EV	3.873
Curvv	3.873
Punch EV	3.869
Tiago EV	3.863
Hexa	3.847
Zest	3.827
Bolt	3.825
Indica	3.824
Winger	3.818
Indigo	3.816
Sierra	3.803
Xenon	3.803
Sumo	3.802